

Classical Demand I: UMP and EMP

Advanced Microeconomics 1: Economic Decision Making. Lecture 2

Luke Heath Milsom

Where we left the question

- **Done:** household inflation $\pi^h = \sum_I b_I^h \pi_I \rightarrow$ who faced what.
- **Done, with one assumption:** basket indexation over-compensates.
- **Missing:** the *right* compensation. “As well off as before” still has no number attached.

Today we cheat: **assume the household's preferences are known**, and construct the ideal answer. In lectures 3–4 we will drop that assumption.

Today's route

1. Preferences becomes numbers: when does $\succsim$ become a **function** u ?
2. Two optimization problems — maximisation (UMP), and minimisation (EMP).
3. EMP produces the cost of living as a mathematical object: $e(p, u)$.
4. From e : the ideal answers to our inflation-compensation question.

From preferences as a ranking to as a function

Standing assumptions on $\succsim$

Definition · monotone / locally nonsatiated

$\succsim$ is...

-**monotone** if $y \gg x \Rightarrow y \succ x$

-**strongly monotone** if $y \geq x, y \neq x \Rightarrow y \succ x$.

-**locally nonsatiated (LNS)** if for every $x \in X$ and every $\varepsilon > 0$ there is $y \in X$ with $\|y - x\| \leq \varepsilon$ and $y \succ x$.

- Monotone implies that more is better.
- LNS is weaker: there is no bliss point nearby.
- LNS is exactly what makes households spend their budget.

Convexity

Definition · convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is convex: $y \succsim x$ and $z \succsim x$ imply $\alpha y + (1 - \alpha)z \succsim x$ for $\alpha \in [0, 1]$.

$\succsim$ is **Strictly convex**: if $\alpha y + (1 - \alpha)z \succ x$ whenever $y \neq z$, $\alpha \in (0, 1)$.

- Ensures a taste for diversification — diminishing marginal rates of substitution.
- Convexity in preferences makes the set of optimal choices convex also.

Why a utility function can fail to exist

Rationality alone is not enough. The classic counterexample, on $X = \mathbb{R}_+^2$:

Definition · lexicographic (dictionary) preferences

$x \succsim y$ if either $x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$.

- Complete, transitive, strongly monotone, strictly convex — everything so far. Yet **no utility function represents it.**
- Why?

Finally, continuity

Definition · continuity

$\succsim$ is **continuous** if for any sequence $\{(x^n, y^n)\}_{n=1}^{\infty}$ with $x^n \succsim y^n$ for all n , $x^n \rightarrow x$ and $y^n \rightarrow y$, we have $x \succsim y$. (Preferences cannot jump at the limit.)

- Lexicographic fails it: $x^n = (\frac{1}{n}, 0) \succsim y^n = (0, 1)$ for every n , yet in the limit $y = (0, 1) \succ (0, 0) = x$.

Debreu's theorem

Theorem · Debreu

If the rational preference relation $\succsim$ on X is continuous, there is a *continuous* utility function $u(x)$ that represents $\succsim$.

Proof: beyond our scope; see MWG 3.C.

Ordinal, not cardinal

- u is not unique: any strictly increasing f makes $v = f(u)$ represent the *same* $\succsim$.
E.g. $x_1^\alpha x_2^{1-\alpha}$ and $\alpha \ln x_1 + (1 - \alpha) \ln x_2$.
- **Levels of u mean nothing.** “Utility fell by 3” doesn’t mean anything. This is why we need more than just u to answer the inflation question.

The consumer's problem

Where we are at

- We see the choices individuals make $x(p, w)$ under given p and w from a possible commodity vector X and given their budget constraint $B(p, w)$.
- Under some assumptions (rationality, continuity, LNS) these choices come from an underlying preference relation $\succsim$ which has a utility representation u .
- We want to know: if p changes how does x change and thus the welfare of consumers.
- To know this we have to formalise the link between p and choices — how does a consumer choose their consumption bundle?

How does a consumer choose their consumption bundle

- We've actually already see the answer: **choose the most preferred affordable bundle**. Pretty obvious.
- Given our notation and assumptions we can now formalise this sentence:

$$\max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

- This is the **utility maximisation problem** (UMP).
- NB: Nobody computes first-order conditions in a supermarket. The claim is that choices are *consistent enough* to be described this way.

The utility maximization problem

Rational, continuous, LNS $\succsim$ on $X = \mathbb{R}_+^L$, represented by continuous u ; prices $p \gg 0$, wealth $w > 0$:

$$(UMP) \quad \max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

Theorem · existence

If $u(\cdot)$ is continuous, the UMP has a solution.

Proof → p. 40

Now we can pull the string and see where the assumptions take us by solving the problem.

What this means for the observed choices $x(p, w)$

Theorem · MWG 3.D.2

If u is a continuous utility function representing an LNS preference relation $\succsim$ on $\mathbb{R}_+^L$, then $x(p, w)$ is the solution to the UMP and thus satisfies...

- (i) **homogeneity of degree zero:** $x(\alpha p, \alpha w) = x(p, w)$ for $\alpha > 0$;
- (ii) **Walras' law:** $p \cdot x = w$ for all $x \in x(p, w)$;
- (iii) **convexity/uniqueness:** convex $\succsim \Rightarrow x(p, w)$ a convex set; strictly convex $\succsim \Rightarrow$ single-valued.

Proof $\rightarrow$ p. 41

In Lecture 1, (i) and (ii) were *assumptions* about data. Now they are consequences.

Solving it: Lagrange

$$\max_{x \geq 0} u(x) + \lambda(w - p \cdot x)$$

- If u is continuously differentiable, an optimum x^* admits a multiplier $\lambda \geq 0$ s.t.

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell, \quad \text{with equality if } x_\ell^* > 0.$$

- Interior optimum: $\frac{\partial u(x^*)/\partial x_\ell}{\partial u(x^*)/\partial x_k} = \frac{p_\ell}{p_k}$. That is, **MRS = price ratio**.

What is λ ?

- λ is the shadow value of money i.e. $\frac{du}{dw} = \lambda$.

- To see this...

- Start: $\frac{du}{dw} = \frac{du(x^*)}{dx^*} \frac{dx^*}{dw}$.

- Sub in FOC ($\frac{du(x^*)}{dx^*} = \lambda p$ to find $\frac{du}{dw} = \lambda p \frac{dx^*}{dw}$.

- From Walrus $p \cdot x^* = w$ differentiate w.r.t. w to find $p \frac{dx^*}{dw} = 1$ sub in, then we have $\frac{du}{dw} = \lambda$ as required.

Worked: Cobb–Douglas

$u(x) = x_1^\alpha x_2^{1-\alpha}$, $\alpha \in (0, 1)$. Interior FOCs + Walras' law give

$$x_1(p, w) = \frac{\alpha w}{p_1}, \quad x_2(p, w) = \frac{(1 - \alpha) w}{p_2}.$$

- **Constant budget shares:** $b_1 = \alpha$, $b_2 = 1 - \alpha$ — exactly the b_i^h of the motivation charts. A Cobb–Douglas household *is* a row of the NBB quartile table.

The indirect utility function

Definition · indirect utility

$v(p, w) = u(x^*)$ for any $x^* \in x(p, w)$: the maximal utility attainable at (p, w) .

Proposition · properties (MWG 3.D.3)

$v(p, w)$ is (i) h.d.0; (ii) increasing in w and nonincreasing in p_ℓ ; (iii) quasiconvex: $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex; (iv) continuous.

Proof deferred to MWG.

“Worse off from the price change” now has a formal translation: $v(p^1, w) < v(p^0, w)$.

Why v cannot be the final answer

- v depends on which representation we picked: replace u by $f(u)$ and every value of v changes. The size of $v(p^1, w) - v(p^0, w)$ is meaningless.
- The key issue is units: Belgium does not index wages in utils, but rather euros.
- The fix is to change the question. Instead of “how much utility from this budget?”, invert: “how much budget for this standard of living?”.

The expenditure minimization problem

The dual problem

- Thus far we have shown that under some assumptions (rationality...) individual choices x are the result of their UMP.
- This gave us $v(p, w)$, the standard of living achievable for any (w, p) measured in utils.
- Utils are not useful units (not even cardinal), to know how much we should compensate, we need to speak in euros.
- The “dual” expenditure minimisation problem (EMP) can help...

The expenditure minimization problem

For $p \gg 0$ and $u > u(0)$:

$$(\text{EMP}) \quad \min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq u.$$

Definition · expenditure function

$e(p, u) = p \cdot x^*$ for any solution x^* : the *minimal cost of reaching the standard of living u at prices p .*

- The inflation problem: $e(p^{2022}, u^{2021})$ is **what it costs in 2022 to be as well off as in 2021**, this is exactly the object we need.
- The *dual* of the UMP: objective and constraint swap roles.

UMP and EMP are the same problem

Theorem · MWG 3.E.1

Under the maintained assumptions (continuity, LNS), for $p \gg 0$:

- (i) x^* optimal in the UMP at wealth $w > 0 \Rightarrow x^*$ optimal in the EMP at required utility $u(x^*)$; the minimized expenditure is exactly w .
- (ii) x^* optimal in the EMP at $u > u(0) \Rightarrow x^*$ optimal in the UMP at wealth $p \cdot x^*$; the maximized utility is exactly u .

Proof → p. 42

So at the optimum $u^* = v(p, w)$ and $w = e(p, u^*)$ so sub one into the other to find...

$$e(p, v(p, w)) \equiv w, \quad v(p, e(p, u)) \equiv u.$$

Properties of the cost of living

Theorem · MWG 3.E.2

$e(p, u)$ is

- (i) homogeneous of degree **one** in p ;
- (ii) strictly increasing in u and nondecreasing in each p_ℓ ;
- (iii) **concave** in p ;
- (iv) continuous in p , for $p \gg 0$.

Proof → p. 43

- (i): double every price and the cost of the old life doubles → wage indexation is the solution? Only for *proportional* inflation.
- (iii) Concavity: as one price rises, cost rises *less than linearly*, because the household substitutes. **Lecture 1's over-compensation result is due to this curvature.**

EMP solution: Hicksian demand

The solution to the EMP is a set of demands for a given (p, u) .

Definition · Hicksian (compensated) demand

$h(p, u) \subseteq \mathbb{R}_+^L$: the solution set of the EMP — the cheapest bundle(s) delivering the standard of living u at prices p .

Theorem · MWG 3.E.3

For $p \gg 0$: (i) **h.d.0 in p** : $h(\alpha p, u) = h(p, u)$; (ii) **no excess utility**: $u(x) = u$ for any $x \in h(p, u)$; (iii) $\text{convex} \rightsquigarrow h(p, u) \text{ convex}$; $\text{strictly convex} \Rightarrow \text{single-valued}$. [Proof → p. 44](#)

Duality identities, from the equivalence theorem:

$$h(p, u) \equiv x(p, e(p, u)), \quad x(p, w) \equiv h(p, v(p, w)).$$

The compensated law of demand

Theorem · MWG 3.E.4

Under the maintained assumptions, with $h(p, u)$ single-valued for $p \gg 0$: for all p' and p'' we have that,

$$(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0.$$

Proof → p. 45

- Own-price version: compensated own-price effects are nonpositive, globally. *Marshallian* demand obeys no such law (Giffen goods for example).

Two compensations

From (p^0, w) with $x^0 = x(p^0, w)$, prices move to p^1 . Two ways to “compensate”:

Slutsky (Lecture 1)

$\Delta w_S = p^1 \cdot x^0 - w$: keep the *old bundle* affordable.

Observable today. **Over-pays** (the L1 theorem). This is what basket indexation does.

Hicksian (today)

$\Delta w_H = e(p^1, u^0) - w$: keep the *old standard of living* reachable.

Exact — the ideal answer. Needs preferences.

$\Delta w_H \leq \Delta w_S$ since $e(p^1, u^0) \leq p^1 \cdot x^0$ (x^0 attains u^0 , so it is feasible in the EMP).

The gap between the columns is the **substitution bias**.

**So, how much should the Belgian government
compensate for inflation?**

Money-metric utility: euros as the unit

Fix any reference prices $\bar{p} \gg 0$ and consider, as a function of (p, w) :

$$e(\bar{p}, v(p, w)).$$

- This is itself an indirect utility function — a strictly increasing transform of v — but its unit is **euros at prices $\bar{p}$** : the budget needed, at reference prices, to live as well as you do at (p, w) .
- Differences of *this* object are meaningful money amounts.

The ideal answers: EV and CV

Let $u^0 = v(p^0, w)$, $u^1 = v(p^1, w)$; and note that $e(p^0, u^0) = e(p^1, u^1) = w$.

Definition · equivalent variation

$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0) = e(p^0, u^1) - w$. The change in spending needed at p^0 to achieve u^1 relative to u^0 .

Definition · compensating variation

$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0) = w - e(p^1, u^0)$. The change in spending needed at p^1 to achieve u^1 relative to u^0 .

The ideal answers: EV and CV

- Both are “ideal”, they differ only in their reference point. Which one policy you should choose is an normative choice.
- For us, the 2022 inflation has happened, we're at p^1 so how large should the compensation be is a CV question: $-CV^h = e^h(p^1, u^0) - w$ is the correct indexation payment for household h .

The true index vs. the published index

Definition · Konüs (true cost-of-living) index

$$P_K(p^0 \rightarrow p^1; u^0) = \frac{e(p^1, u^0)}{e(p^0, u^0)} \quad \text{vs. Laspeyres:} \quad P_L = \frac{p^1 \cdot x^0}{p^0 \cdot x^0}.$$

Proposition · Laspeyres bounds the truth from above

$$P_K \leq P_L. \text{ Proof: } x^0 \text{ attains } u^0, \text{ so } e(p^1, u^0) \leq p^1 \cdot x^0; \text{ and } e(p^0, u^0) = p^0 \cdot x^0 = w. \square$$

- The Belgian gov. mechanically over-state the rise in the cost of living.
- Next question: By how much? We can answer this now if we assume a utility function.

The exact bill under Cobb–Douglas

Cobb–Douglas over (energy, other), energy share α : $u(x) = x_E^\alpha x_O^{1-\alpha}$ gives:

$$e(p, u) = u \left(\frac{p_E}{\alpha} \right)^\alpha \left(\frac{p_O}{1-\alpha} \right)^{1-\alpha}.$$

The exact compensation is $-CV = e(p^1, u^0) - w$, with $w = e(p^0, u^0)$. Form the ratio — only p_E moves, p_O and u^0 are fixed, so every factor cancels except the energy term:

$$\frac{e(p^1, u^0)}{e(p^0, u^0)} = \frac{\cancel{u^0} (p_E^1/\alpha)^\alpha (\cancel{p_O/(1-\alpha)})^{1-\alpha}}{\cancel{u^0} (p_E^0/\alpha)^\alpha (\cancel{p_O/(1-\alpha)})^{1-\alpha}} = \left(\frac{p_E^1}{p_E^0} \right)^\alpha = (1+g)^\alpha, \quad g \equiv \frac{p_E^1}{p_E^0} - 1.$$

So the exact bill is $\frac{-CV}{w} = (1+g)^\alpha - 1$. Expand for the intuition (Taylor in g):

$$(1+g)^\alpha - 1 = \underbrace{\alpha g}_{\text{share} \times \text{price rise}} - \underbrace{\frac{1}{2} \alpha(1-\alpha) g^2}_{\text{substitution saving}} + \dots$$

Two biases: substitution, then heterogeneity

Home energy rose $g = 57.9\%$ in 2022 (NBB).

Bias 1 — substitution (each paid its *own* basket αg , vs its true bill):

	true bill $(1 + g)^\alpha - 1$	own basket αg	over-pay
poor, $\alpha = 0.10$	4.7%	5.8%	1.1 pp
rich, $\alpha = 0.06$	2.8%	3.5%	0.7 pp

Bias 2 — one basket for all the gov. pays a single $\bar{s} g$, with $\bar{s} = 6.4\%$, the home-energy weight in Statbel's 2022 CPI basket:

	true bill $(1 + g)^\alpha - 1$	actual index $\bar{s} g$	index – bill
poor, $\alpha = 0.10$	4.7%	3.7%	–1.0 pp
rich, $\alpha = 0.06$	2.8%	3.7%	+0.9 pp

So we're done here right?

- We've answered our questions: Who was hurt by the 2022 inflation, and how could the gov. compensate them.
- But in doing so we have made some unsubstantiated assumptions:
 - We assumed a u .
 - We assumed rationality.
 - We assumed the change in p was one-off and known.

So we're done here right?

- We've answered our questions: Who was hurt by the 2022 inflation, and how could the gov. compensate them.
- But in doing so we have made some unsubstantiated assumptions:
 - We assumed a u . In L3 we will instead go from x to e .
 - We assumed rationality. In L4 we will stress-test this assumption.
 - We assumed the change in p was one-off and known. In L7+ we will consider uncertainty and risk.

Where the question stands

Takeaways

- **The ideal answer exists:** $e(p^1, u^0) - w$, per household — “as well off as before”.
- **Upgrades:** h.d.0 and Walras’ law are theorems under the maintained hypothesis, not assumptions about data.
- **The catch:** everything today was built from u — which no dataset contains, we had to assume it.
- **Next:** Lecture 3 connects e , h and v to observable demand.

Appendix: proofs

Proof: the UMP has a solution

- The budget set $B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$ is **bounded**: for each ℓ , $x_\ell \leq w/p_\ell$ for every $x \in B_{p,w}$ (this uses $p \gg 0$).
- It is **closed**: an intersection of closed half-spaces with the closed orthant.
- Closed and bounded in $\mathbb{R}^L$ = **compact**; and $B_{p,w} \neq \emptyset$ since $0 \in B_{p,w}$. A continuous function attains a maximum on a nonempty compact set. ■

Proof: properties of Walrasian demand (MWG 3.D.2)

- (i) *h.d.o.* $B_{\alpha p, \alpha w} = \{x : \alpha p \cdot x \leq \alpha w\} = B_{p, w}$: same constraint set, same objective, hence the same maximizers.
- (ii) *Walras' law*. Suppose $x \in x(p, w)$ with $p \cdot x < w$. Then a small ball around x lies inside the budget set, and LNS supplies y in that ball with $y \succ x$: an affordable strict improvement, contradicting optimality. So $p \cdot x = w$.
- (iii) *Convexity/uniqueness*. Let $x, x' \in x(p, w)$, both with utility u^* . The mixture $x'' = \alpha x + (1 - \alpha)x'$ is affordable ($B_{p, w}$ convex) and quasiconcavity gives $u(x'') \geq u^*$, so $x'' \in x(p, w)$: the solution set is convex. If u is *strictly* quasiconcave and $x \neq x'$, then $u(x'') > u^*$ — contradicting maximality of u^* . So the solution is unique. ■

Proof: UMP $\iff$ EMP (MWG 3.E.1)

(i) Let x^* solve the UMP at wealth w ; by Walras' law, $p \cdot x^* = w$. Suppose x^* does *not* solve the EMP at utility $u(x^*)$: some x' has $u(x') \geq u(x^*)$ and $p \cdot x' < w$. By LNS there is x'' near x' with $u(x'') > u(x') \geq u(x^*)$ and still $p \cdot x'' < w$ — an affordable bundle beating x^* in the UMP. Contradiction. The minimized expenditure is $p \cdot x^* = w$.

(ii) Let x^* solve the EMP at $u > u(0)$; then $x^* \neq 0$, so $w' := p \cdot x^* > 0$. Suppose x^* does *not* solve the UMP at w' : some x' has $p \cdot x' \leq w'$ and $u(x') > u(x^*) \geq u$. By continuity, $\alpha x'$ with $\alpha < 1$ close to 1 still has $u(\alpha x') > u$, at cost $\alpha p \cdot x' < w'$ — cheaper and feasible in the EMP, contradicting minimality. Finally, $u(x^*) = u$ exactly: if $u(x^*) > u$, the same scaling applied to x^* itself yields a cheaper feasible bundle. ■

Proof: properties of the expenditure function (MWG 3.E.2)

- (i) *h.d. 1.* The constraint set $\{x : u(x) \geq u\}$ does not involve p ; scaling $p \rightarrow \alpha p$ scales the objective, hence the minimized value, by α .
- (ii) *Strictly increasing in u .* Suppose $u' > u$ but $e(p, u') \leq e(p, u)$, and let x' attain u' at cost $e(p, u')$. By continuity, $\alpha x'$ with $\alpha < 1$ near 1 still gives utility $> u$, at cost $\alpha e(p, u') < e(p, u)$: contradicts the definition of $e(p, u)$. (Nondecreasing in p_ℓ : raising p_ℓ weakly raises the cost of every feasible bundle.)
- (iii) *Concavity.* Fix $\bar{u}$; let $p'' = \alpha p + (1 - \alpha)p'$ and let x'' be optimal at p'' . Then x'' attains $\bar{u}$, hence is *feasible* (not necessarily optimal) at prices p and at prices p' :

$$e(p'', \bar{u}) = p'' \cdot x'' = \alpha p \cdot x'' + (1 - \alpha) p' \cdot x'' \geq \alpha e(p, \bar{u}) + (1 - \alpha) e(p', \bar{u}). \quad \blacksquare$$

(iv) Continuity: the maximum theorem; see MWG.

Proof: properties of Hicksian demand (MWG 3.E.3)

- (i) *h.d.0 in p*. Minimizing $\alpha p \cdot x$ and minimizing $p \cdot x$ over the same constraint set $\{x : u(x) \geq u\}$ have the same solutions.
- (ii) *No excess utility*. Suppose $x \in h(p, u)$ with $u(x) > u$. Since $u > u(0)$, $x \neq 0$, so $p \cdot x > 0$. By continuity, αx with $\alpha < 1$ near 1 still has $u(\alpha x) > u$ — feasible and strictly cheaper. Contradiction; so $u(x) = u$.
- (iii) *Convexity/uniqueness*. The constraint set $\{x : u(x) \geq u\}$ is convex when u is quasiconcave, and the objective is linear: two minimizers cost the same, and any mixture of them is feasible and costs the same — the solution set is convex. Under *strict* convexity, the mixture of two distinct minimizers lies strictly inside the upper contour set, so a small radial contraction stays feasible and cuts cost — forcing uniqueness. ■

Proof: the compensated law of demand (MWG 3.E.4)

Fix u . Each of $h(p', u)$ and $h(p'', u)$ attains utility u , so each is *feasible* in the other's problem. Cost-minimality therefore gives two inequalities:

$$\begin{aligned} p'' \cdot h(p'', u) &\leq p'' \cdot h(p', u) && (h(p', u) \text{ feasible at prices } p''), \\ p' \cdot h(p', u) &\leq p' \cdot h(p'', u) && (h(p'', u) \text{ feasible at prices } p'). \end{aligned}$$

Add the first to minus the second:

$$(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0. \blacksquare$$